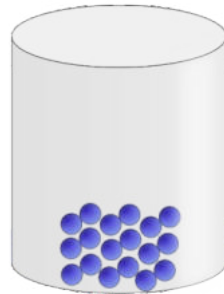
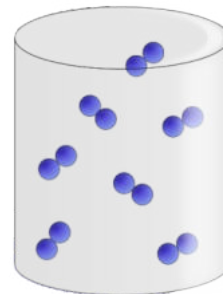


UNIT 07

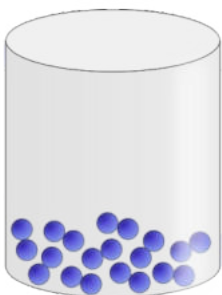
PROPERTIES OF MATTER



Solid



Gas



Liquid

ARTIFICIAL SATELLITES

Matter can be defined in simpler terms as anything that occupies space, has a mass and volume. Everything we see in front of us is composed of matter, even the air we breathe is matter.



EXAMPLES

A bicycle, an apple, an aircraft, the sun etc.

STATE OF MATTER

The state of matter refers to the physical form of a substance and can be classified into three main categories: solid, liquid, and gas.

SOLID

- ▶ Has a definite shape and volume?
- ▶ Particles are tightly packed together and vibrate in a fixed position.
- ▶ Cannot be compressed easily.



LIQUID

- ▶ Has a definite volume but not a definite shape?
- ▶ Particles are loosely packed together and move around freely.
- ▶ Flows to fill any container.



GAS

- ▶ Has no definite shape or volume?
- ▶ Particles are spread out and move rapidly in all directions.
- ▶ Can be compressed easily.



STATES OF MATTER	SHAPE	VOLUME	DENSITY	COMPRESSIBILITY
Solid	Fixed	Fixed	High	Incompressible
Liquid	Not Fixed	Fixed	High	Incompressible
Gas	Not Fixed	Not Fixed	Low	Compressible

CHANGE IN STATE OF MATTER

It is a physical process that occurs when matter is subjected to a change in temperature, pressure, or both.

1. SOLID TO LIQUID (MELTING)

Occurs when the temperature of a solid is increased to its melting point, causing the particles to gain enough energy to overcome their intermolecular forces and move freely, forming a liquid.

2. LIQUID TO SOLID (FREEZING)

Occurs when the temperature of a liquid is decreased to its freezing point, causing the particles to lose energy and become more ordered, forming a solid.

3. Liquid to Gas (Evaporation):

Occurs when the temperature of a liquid is increased, causing some particles to gain enough energy to escape into the air as a gas.

4. GAS TO LIQUID (CONDENSATION)

Occurs when the temperature of a gas is decreased, causing the particles to lose energy and become more ordered, forming a liquid.

5. SOLID TO GAS (SUBLIMATION)

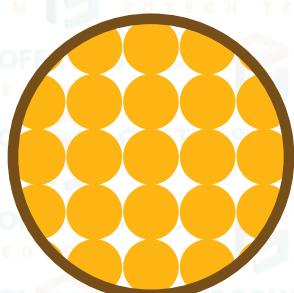
Occurs when a solid is subjected to high temperature and pressure, causing the particles to gain enough energy to escape into the air as a gas. Overall, the change in state of matter is a fundamental concept in physics that helps us understand the physical properties of substances and how they respond to different environmental conditions.

KINETIC MOLECULAR MODEL OF MATTER

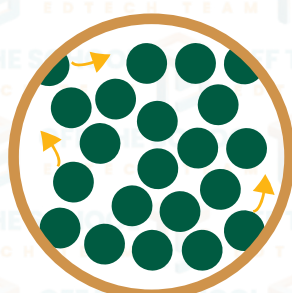
The kinetic molecular model of matter is, Matter is made up of tiny particles called atoms, or group of atoms called molecules. These molecules are always in continuous random motion. According to this model particles are in continuous motion. The evidence of molecular motion is Brownian motion.

BROWNIAN MOTION

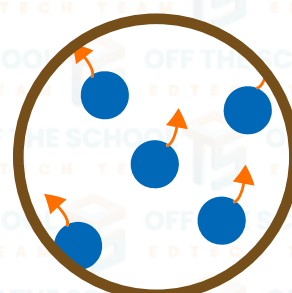
Brownian motion, also known as the random motion of particles, is a phenomenon that was first observed by Scottish *Botanist Robert Brown* in **1827**. In a nutshell, it refers to the seemingly random movement of small particles suspended in a liquid or gas, caused by the collisions of these particles with the rapidly moving molecules of the surrounding fluid. Think of a single particle floating in a glass of water. It would move randomly, bouncing and jostling around due to the constant impacts from the water molecules. This same principle applies to a group of particles, where the overall movement of the group appears random, but is actually the result of many individual collisions. This concept has important applications in fields such as physics, chemistry.



SOLID



LIQUID



GAS

THE MOVEMENT OF PARTICLES IN SOLID HAVE FOLLOWING FEATURES

Brownian motion in solid state refers to the random movement of particles within a solid material. This movement is caused by the collision of the particles with other particles and with the surrounding environment. Students learn about the particle nature of matter and how it affects the physical properties of substances. Brownian motion is an important concept in this area because it helps explain the thermal conductivity, viscosity, and diffusion of solids.

SOME KEY NOTES ON BROWNIAN MOTION IN SOLID STATE INCLUDE

1. Brownian motion is not limited to liquids and gases. It also occurs in solids, but at a much slower rate due to the increased density and rigidity of the material.
 2. Brownian motion is a direct result of the thermal energy in a material. As temperature increases, so does the rate of Brownian motion.
 3. The movement of particles in a solid due to Brownian motion can affect the physical properties of the material. For example, it can lead to an increase in thermal conductivity and viscosity, and contribute to the diffusion of particles through the material.
 4. Brownian motion is important for many real-world applications, such as the study of materials and the behavior of substances at high temperatures.
- Overall, Brownian motion in solid state is a fundamental concept in the study of materials science and thermodynamics. Understanding this phenomenon is crucial for students in grade 9 as they continue to explore the properties of matter and their applications.

THE MOVEMENT OF PARTICLES IN LIQUIDS HAVE FOLLOWING FEATURES

Here are some key points to remember about Brownian motion in a liquid state:

1. Particles in a liquid are in constant random motion due to the collisions they experience with the molecules of the liquid.
2. The movement of these particles is not in a specific direction and is completely random.
3. The movement of particles in a liquid is more rapid at higher temperatures, as the increased energy of the particles leads to more frequent collisions.
4. Brownian motion is a result of the thermal motion of the particles, which is caused by the transfer of energy from the surrounding liquid to the particles.
5. The movement of particles in a liquid can be seen under a microscope, and is often used to study the properties of liquids and the interactions between particles.
6. Brownian motion plays an important role in several fields, including biology (e.g. the movement of cells in a liquid), chemistry (e.g. the diffusion of molecules in a solution), and physics (e.g. the behavior of particles in a liquid).
7. The theory of Brownian motion has been developed and refined over the years and is now considered to be a fundamental aspect of the study of liquids and their properties.

THE MOVEMENT OF PARTICLES IN GASES HAVE FOLLOWING FEATURES

Here are some key points to remember about Brownian motion in a Gaseous state: Brownian motion refers to the random movement of particles in a fluid.

1. In a gas, the particles are in constant motion, colliding with each other and bouncing around randomly.
2. This random movement is due to the thermal energy of the particles, which makes them move with a high velocity.

3. The movement of gas particles in Brownian motion can be described mathematically using the laws of thermodynamics and the kinetic theory of gases.
4. Brownian motion can also be observed under a microscope as the random movement of particles suspended in a fluid.
5. The random motion of gas particles has important implications in fields such as chemistry and physics. For example, it can impact the rate of chemical reactions, the behavior of gases in containers, and the transfer of heat.
6. Brownian motion is named after the Scottish botanist Robert Brown, who first observed the phenomenon in 1827 while studying pollen particles suspended in water.

FORCE AND KINETIC THEORY

Forces and kinetic theory are important concepts in physics, particularly in the study of motion and the behavior of gases.

FORCES

A force is a push or pull upon an object resulting from the object's interaction with another object. Forces can cause an object to accelerate, change direction, or stay in its current state of motion. There are many types of forces, including gravitational forces, electromagnetic forces, and the strong and weak nuclear forces.

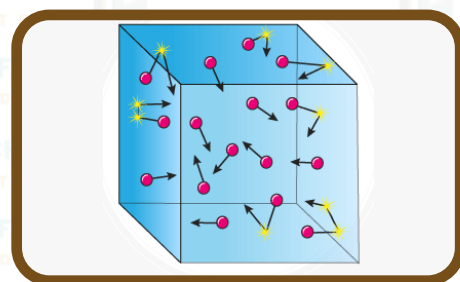
KINETIC THEORY

Kinetic theory is a theoretical framework used to describe and explain the motion of particles in a gas. According to kinetic theory, the behavior of a gas is the result of the constant, random motion of its particles. The theory describes how the temperature, pressure, and volume of a gas are related to the kinetic energy of its particles. Kinetic theory also provides a basis for understanding the behavior of gases under different conditions and is an important tool for understanding many processes in nature, such as weather patterns and the behavior of stars. Students are typically introduced to these concepts as part of an introduction to physics. They will learn the basic principles of forces and motion and how they can be described and analyzed using mathematical equations. They will also learn about the properties of gases and how they can be understood using kinetic theory.

GASES AND KINETIC THEORY

The kinetic theory of gases is a theory that explains the behavior of gases in terms of the motion of their individual molecules. The theory is based on the following assumptions:

1. Gas molecules are in constant random motion, colliding with each other and with the walls of the container.
2. Gas molecules are considered to be point particles, meaning they have no volume and occupy no space.
3. Gas molecules have no intermolecular forces (i.e., they don't attract or repel each other).
4. Gas molecules have a certain average kinetic energy, determined by the temperature of the gas. Using these assumptions, the kinetic theory of gases can be used to explain several gas laws, such as Boyle's law and Charles's law. These laws describe the relationship between the pressure, volume, temperature, and number of molecules in a gas. The kinetic theory of gases



Also helps explain the behavior of gases under different conditions, such as in a piston, nozzle. Overall, the kinetic theory of gases is a useful tool for understanding the behavior of gases and how they interact with their environment.

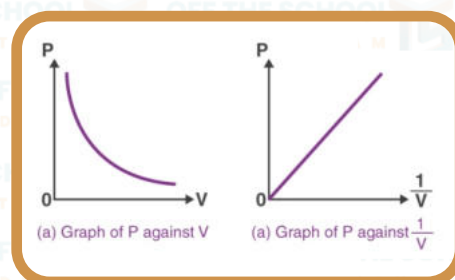
BOYLE'S LAW

Boyle's Law is a scientific law that describes the relationship between pressure and volume of a gas. It states that, for a fixed amount of gas at a constant temperature, the pressure and volume of the gas are inversely proportional. This means that if the volume of a gas increases, its pressure decreases, and vice versa.

Mathematically, Boyle's Law can be expressed as:

$$P_1 \times V_1 = P_2 \times V_2$$

Where P_1 and V_1 are the initial pressure and volume of the gas, and P_2 and V_2 are the final pressure and volume of the gas.



APPLICATIONS

Boyle's Law is an important concept in physics and is widely used in many applications, such as in the design of engines, air conditioning systems, and medical equipment. It is a fundamental principle that helps us understand how gases behave and is a key aspect of many branches of physics, including thermodynamics and fluid mechanics.

CHARLES'S LAW

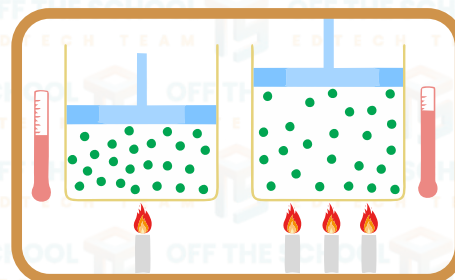
Charles' Law, also known as the Law of Volume-Temperature states that the volume of a gas is directly proportional to its absolute temperature, provided the pressure remains constant. In mathematical terms, it can be expressed as:

$$\frac{V}{T} = k$$

Where V is the volume of the gas, T is the temperature in kelvins, and k is a constant.

This means that if we increase the temperature of a gas, its volume will increase, and if we decrease the temperature of a gas, its volume will decrease. Charles' Law is one of the gas laws, which describe the behavior of gases under different conditions.

These laws help to explain the behavior of gases in a variety of real-world applications, such as weather forecasting, atmospheric science, and engineering. It is important to note that Charles' Law only applies to ideal gases, which are gases that behave as if they were made up of a large number of tiny, non-interacting particles. In reality, gases do not behave exactly as ideal gases, but Charles' Law provides a good approximation for most real-world gases under normal conditions.



We cannot solve our problems with the same thinking we used when we created them.

Albert Einstein

